Optima Smart Vision Frame

Hardware & System Architecture Specification



The **Optima Smart Vision Frame** is an assistive wearable device engineered to grant visually impaired users real-time context regarding their environment. Featuring dual high-resolution micro-cameras positioned at the far-left and far-right extremities of the front chassis, Optima captures wide-angle stereoscopic vision. The frame pairs seamlessly via dual Bluetooth 5.3 LE and Wi-Fi Direct protocols to a host mobile application powered by the Gemma 4 / Gemini multimodal AI engine.

1. FRONT-FACING DUAL CAMERA ARCHITECTURE

Extreme Outer Position

Mounting optics at the far front corners mimics natural human eye alignment, preventing hand or frame obstruction during reading and object interaction.

Stereoscopic FOV & Depth

Dual 120° wide-angle sensors capture overlapping visual fields, enabling accurate spatial geometry mapping and depth assessment for complex scenes.

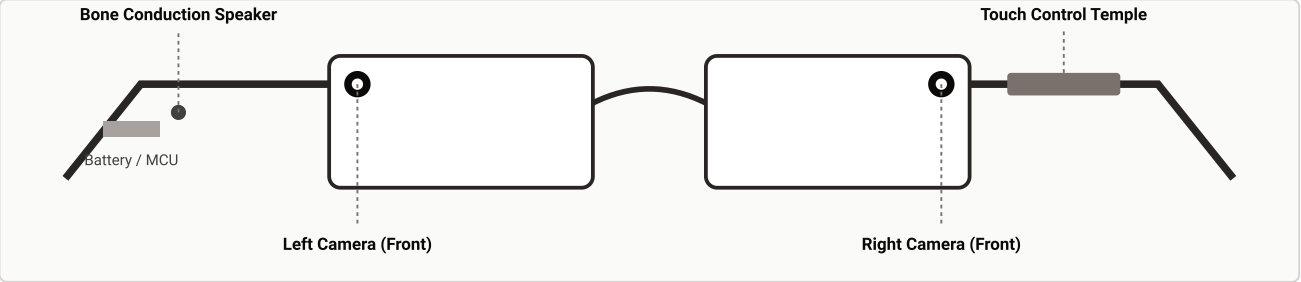
Low-Latency Capture

Hardware trigger button or tactile tap captures full 1080p uncompressed frames instantly, buffering locally for rapid wireless uplink.

Multi-Angle OCR Accuracy

Stereo perspective captures printed documents and signage from multiple angles simultaneously, eliminating distortion from curved pages or skewed angles.

2. HARDWARE COMPONENT SCHEMATIC



3. WIRELESS & MOBILE CONNECTIVITY

Optima operates using a hybrid wireless bridge. Bluetooth 5.3 Low Energy maintains continuous control signaling, touch button inputs, and telemetry. When an action trigger occurs (e.g., Read or Describe), a dedicated high-speed Wi-Fi Direct connection transmits image frames directly to the connected mobile phone within 120 milliseconds.

4. PHYSICAL CONTROLS & INTERACTION MAPPING

Gesture / Input	Hardware Action	Triggered Function
Single Tap (Right Temple)	Instant Frame Capture	Read text in front of user
Double Tap	Stereoscopic Scene Snapshot	Describe surrounding environment
Swipe Forward	Audio Control	Replay last spoken answer
Swipe Backward	Audio Control	Stop active speech playback
Press & Hold (2 sec)	Mic Activation	Voice prompt / Custom question

5. MOBILE APP PROCESSING WORKFLOW

The Optima hardware operates in tandem with the Optima Mobile Companion app running on iOS/Android. Processing is executed via the Gemma 4 / Gemini multimodal pipeline.

01 **User Trigger:** User taps the physical right temple sensor or issues a voice command. Dual front extremity cameras instantly capture stereoscopic image payload.

02 **High-Speed Uplink:** Image payload is sent via Wi-Fi Direct to the Optima Mobile App. The app validates image resolution and contrast.

03 **Multimodal AI Pipeline:** The mobile backend issues an HTTP POST request to `/analyze` with payload parameters `{ image_base64, action, question, language }`.

04 **Gemma 4 Inference:** The image and prompt are parsed by Gemma 4 (e.g. `gemma-4-26b-a4b-it`). Depending on the mode selected, the model yields standard text for speech synthesis:

- **Read:** Extract and structure visible text.
- **Describe:** Outline spatial positioning of objects.
- **Translate:** Convert foreign text into local spoken language.
- **Explain:** Simplify complex charts, diagrams, or notes.

05 **Real-Time Audio Delivery:** Synthesized speech streams back to the Optima frame speakers via Bluetooth audio profile (A2DP) with instant Replay/Stop control available on the temple touchpad.

6. SAFETY & USAGE BOUNDARIES

Optima is strictly engineered as a reading, scene understanding, and contextual assistance tool. Due to network transmission latency (approx. 1.2–2.5 seconds end-to-end) and potential vision model misinterpretations, the device must not be used as an active real-time navigation or mobility safety device in hazardous environments.